Examiner only

4. *(a)* Heat is supplied to an ideal gas in a sealed container of fixed volume. Explain carefully the physical processes that cause an increase in the **temperature and pressure** of the gas. [6 QER]

Examiner only

(b) A container of fixed volume contains oxygen gas at a temperature of 293 K.

(i) Five oxygen molecules have speeds 400, 425, 450, 550 and 625 m s^{-1}. Determine their rms speed. [2]

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(ii) Using an appropriate calculation, determine whether or not the rms speed calculated in part *(b)*(i) is consistent with the expected rms speed of the molecules of the gas at this temperature. (Relative molecular mass of oxygen gas = 32.) [3]

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(iii) If the gas in the container is heated and the pressure of the gas increases by 20% of its initial value, determine the rms speed of the molecules of the gas. [2]

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